

JULIAN ZHANG

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EDUCATION

University of Waterloo

(Sep 2024 - Apr 2029)

Bachelor of Computer Science and Finance Double Major (CFM) | *President's Scholarship of Distinction*

EXPERIENCE

Oakville Centre For Vision - Software Engineer

(May 2024 - Sep 2024)

- Created a web platform using **C# Monogame** and **Python** for eye-tracking goggles to analyze vision patterns in professional athletes. Collaborated directly with optometrist Dr. Ronald Gall
- Focused on data analysis of **8+ tests** that examine gaze vectors for saccades, fixations, and pursuits
- Collaborated with a hardware team to modernize a **6,000+ line** codebase for improved functionality

University of Toronto Clean Energy Lab - Research Assistant

(Jun 2022 - Sep 2023)

- Applied Crystal Graph Convolutional Neural Network (CGCNN) to examine trends in chemical properties
- Analyzed results for material discovery with a focus on advancing lithium-ion alternatives for sustainable energy production. Implemented **PyTorch** regression models achieving an accuracy of **97%**
- Presented findings to Prof. Oleksandr Voznyy and PhD Students as the group's youngest member

Team Canada Debate - Member

(May 2023 - Aug 2024)

- Represented Canada as **1 of 12** debaters at the 2024 Belgrade World School Debating Championship (WSDC), honing advanced critical analysis, structured thinking, and public speaking skills
- Conducted research across politics, economics, and international relations, engaging with field experts
- Organized 5+ debate tournaments attracting **500+** participants and raising **\$10,000+** for charity

PROJECTS

SignWave | OpenCV, Flask, MediaPipe, Whisper

- AI translator converting text and audio inputs into American Sign Language animations, improving accessibility for the deaf and hard of hearing. **First Place and Best Use of GitHub @ JamHacks VII**

EdgeFund | Yahoo Finance, Pandas, NumPy, matplotlib

- Digital stock broker statistically selecting a stock portfolio, outperforms the TSX/S&P by **30% in 1 year**

Personal Website | HTML/CSS, TypeScript, Node.js, Quartz

- Website featuring a portfolio and course notes, supporting **100+ students** with educational content

SKILLS

- Languages:** C/C++/C#, Python, Racket, Java, JavaScript, HTML/CSS, LaTeX
- Libraries:** React.js, Node.js, NumPy, matplotlib, Pandas, PyTorch, TensorFlow, Yahoo Finance
- Coursework:** Algorithm Design and Data Abstraction, Financial Markets and Data Analytics
- Certifications:** Supervised Machine Learning, Advanced Learning Algorithms, Unsupervised Learning

SELECTED AWARDS

- Canadian National British Parliamentary Debate Championship - **1st speaker & team**
- Munk Debates ReSolved Global Debate Competition - **Champion, awarded \$5,000**
- JamHacks VII Hackathon (300+ competitors) - **First Place Overall, Best Use of GitHub**
- American Invitational Math Exam (AIME) Qualifier - **Top 8% Worldwide**